# CSE4227 Digital Image Processing

## Lecture 02 – Digital image Fundamentals

Dr. Kazi A Kalpoma

Professor, Department of CSE

& Director of ICT

Ahsanullah University of Science & Technology (AUST)

Contact: kalpoma@aust.com

# Today's Contents

- **Image Sensing and Acquisition**
  - **Visual Perception**
  - **Image Formation**
  - **Image Acquisition**
  - **Image formation model**

Chapter 2 from R.C. Gonzalez and R.E. Woods, Digital Image Processing (3rd Edition), Prentice Hall, 2008 [ **Section 2.1, 2.2, 2.3** ]

# What is meant by visual perception?

- **Visual perception** is the ability **to see** and **interpret** (analyze and give meaning to) the **visual** information that surrounds us.

- There is a large difference between the image we display and the image we actually perceive.

# Human Visual System

- There is a difference between the **luminance** of a pixel on a computer screen and the perceived brightness of this pixel.
- If we **double the screen luminance**, this **does not imply** that the perceived brightness is doubled too.
- The brightness also depends on other factors, such as **contrast around the pixel** and various **cognitive processes**

# Human Visual System

- The human visual system can perform a number of image processing tasks in a manner vastly superior to anything we are presently able to do with computers.

- The best vision model we have!

- If we want to mimic such processing, we need to carefully study the way our eyes and brain do this.

# Human Visual System

- The human visual system consists of two functional parts:
  - **eye and**
  - **(part of the) brain.**

- The brain does all of the complex image processing, while the eye functions as the biological equivalent of a camera.

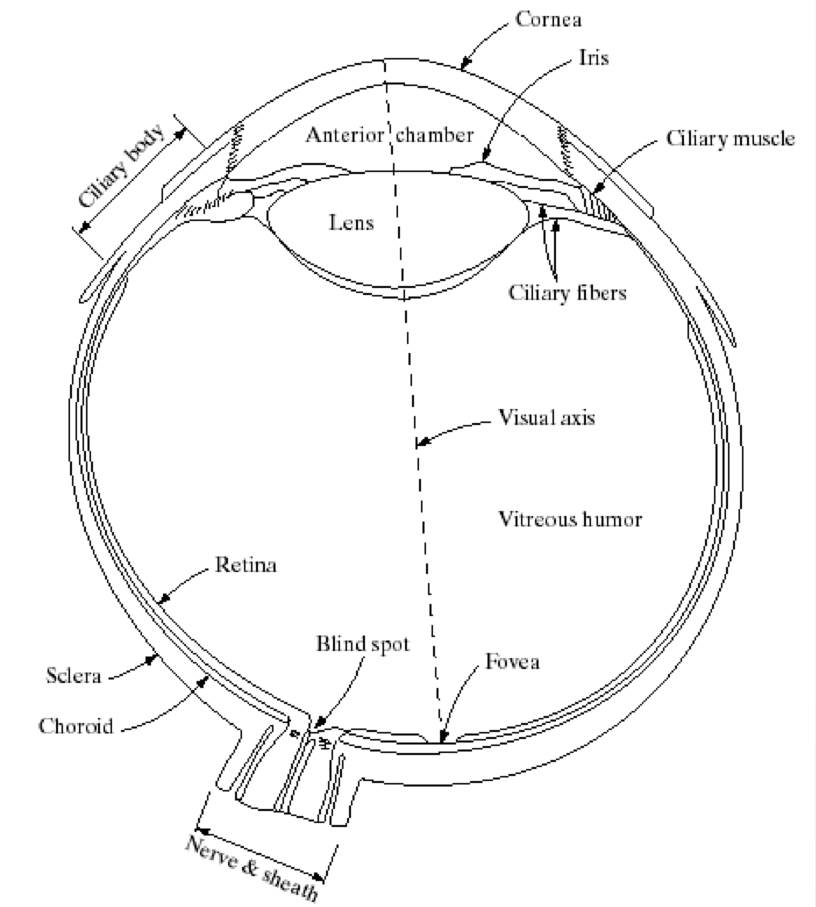
# Structure Of The Human Eye

- The lens focuses light from objects onto the retina

- The retina is covered with light receptors called *cones* (6-7 million) and *rods* (75-150 million)

- Cones are concentrated around the fovea and are very sensitive to colour

- Rods are more spread out and are sensitive to low levels of illumination



- When a light ray hits the eye, it will **first pass through the cornea, then** subsequently through the aqueous humor, the iris, the lens, and the vitreous humor before **finally reaching the retina**.
- The **cornea** is a transparent protective layer, which acts as a lens and **refracts the light**.
- The **iris** forms a round aperture that can vary in size and so **determines the amount of light** that can pass through.
- In the **retina**, the light rays are detected and **converted to electrical signals** by photoreceptors.

# Visual Perception: Human Eye

- The ***lens*** contains 60-70% water, 6% of fat.

- The ***iris*** diaphragm controls amount of light that enters the eye.

- ***Light receptors*** in the ***retina are Cones and Rods:***

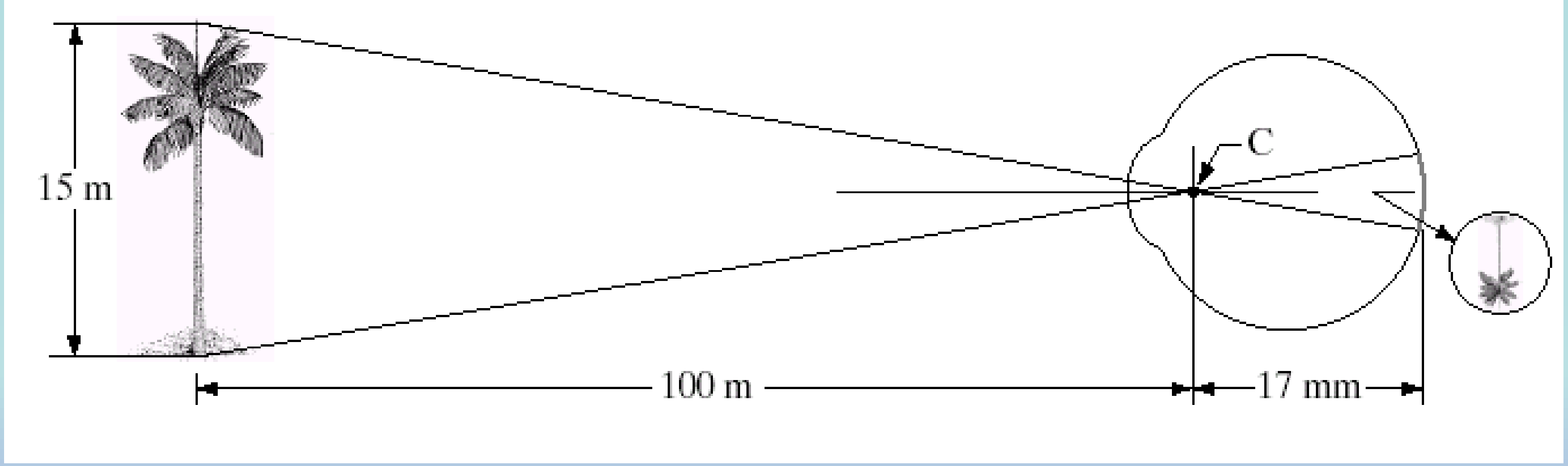
  - ***Cones*** for bright light vision called ***photopic***
    - Cones involve in color vision.
    - Cones are concentrated in ***fovea*** about 1.5x1.5 $mm^2$.

  - ***Rods*** for dim light vision called ***scotopic***
    - Rods are sensitive to low level of light and
    - are not involved in color vision.

- ***Blind spot*** is the region of emergence of the optic nerve from the eye. In this area receptors are absent and no image detection.

# Image Formation In The Eye

- Muscles within the eye can be used to change the shape of the lens allowing us focus on objects that are near or far away
- An image is focused onto the retina causing rods and cones to become excited which ultimately send signals to the brain




**(Height of the tree in the retinal image ) / 17 = (15/100 )**

# Some psychophysics:

## Blind-Spot Experiment

- Draw an image similar to that below on a piece of paper (the dot and cross are about 6 inches apart)

- Close your right eye and focus on the cross with your left eye
- Hold the image about 20 inches away from your face and move it slowly towards you
- The dot should disappear!

# Brightness Adaptation & Discrimination

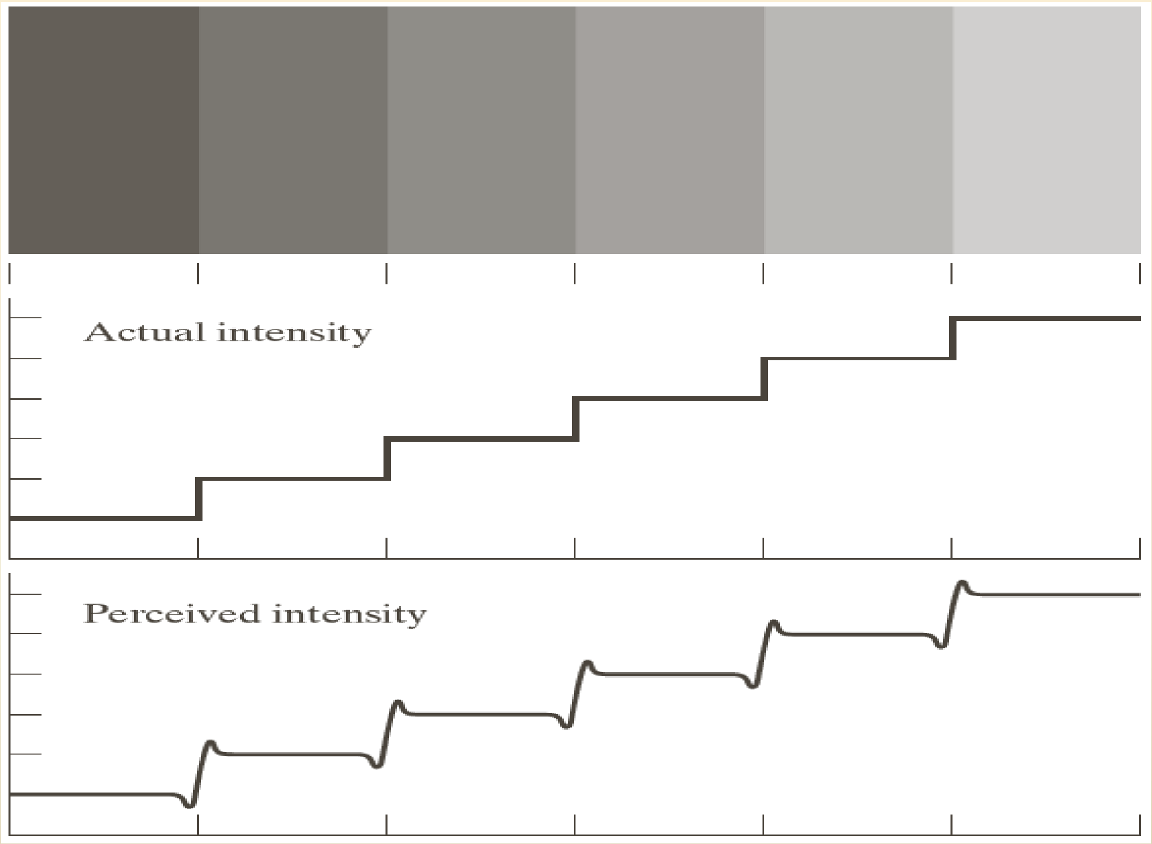
- The human visual system can perceive approximately $10^{10}$ different light intensity levels

- However, at any one time we can only discriminate between a much smaller number – *brightness adaptation*

- Similarly, the *perceived intensity* of a region is related to the light intensities of the regions surrounding it

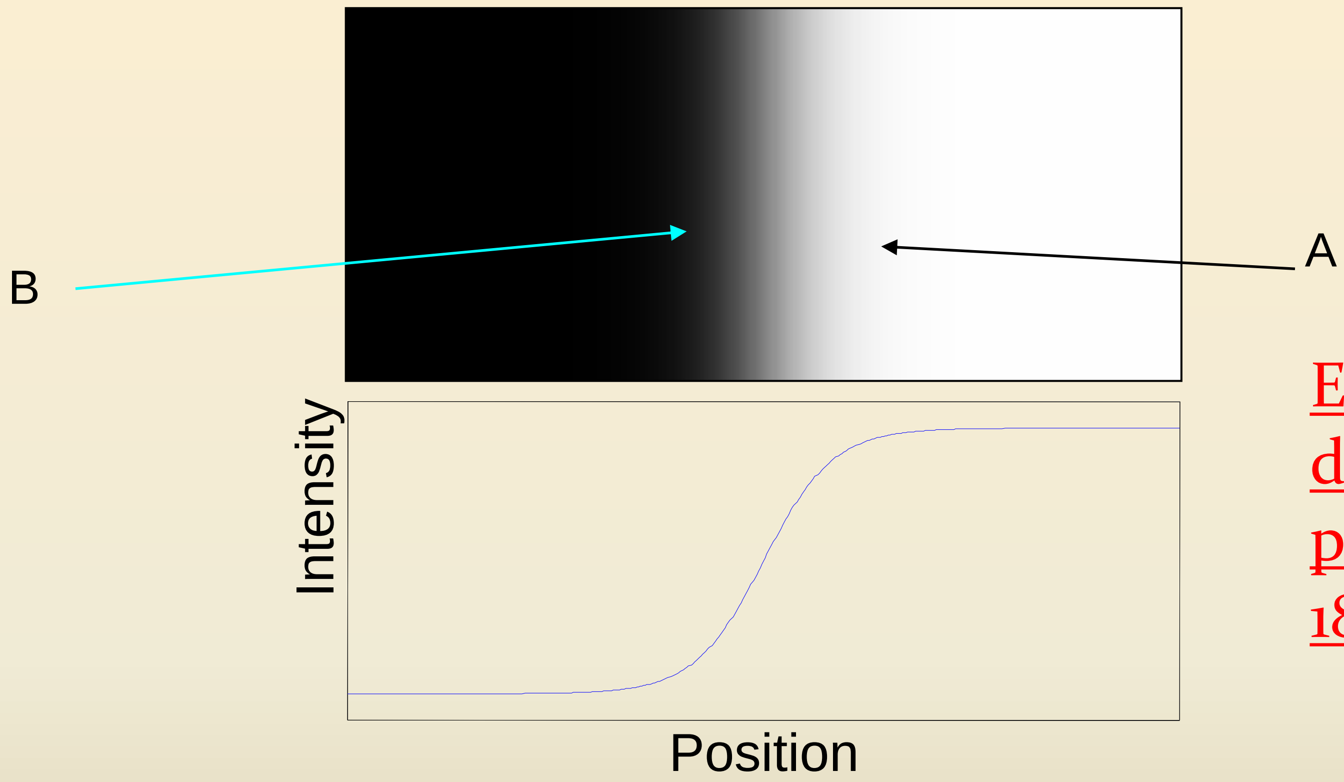
# Brightness Adaptation & Discrimination

**Intensities of surrounding points effect perceived brightness at each point.**




**In this image, edges between bars appear brighter on the right side and darker on the left side.**

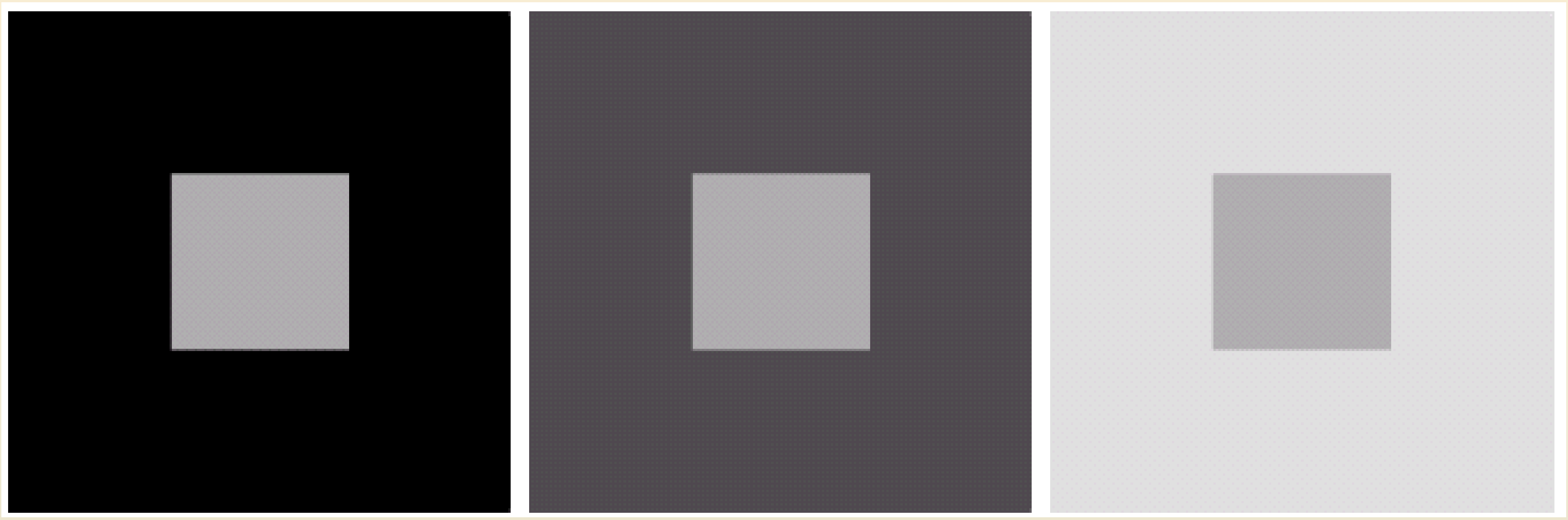
# Mach Band Effect (Cont)




Ernst Mach described this phenomenon in 1865

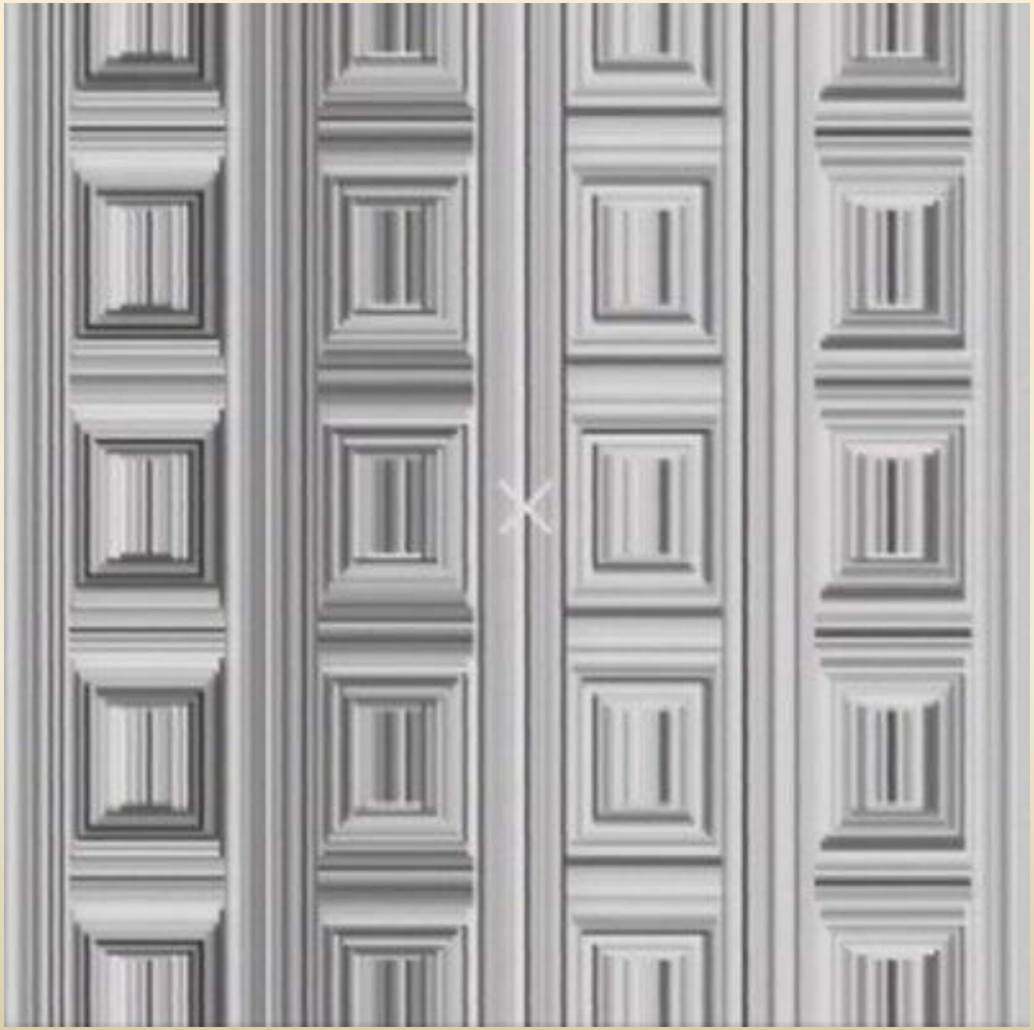
- In area A, brightness perceived is darker while in area B is brighter.
- This phenomenon is called ***Mach Band Effect***.

# Brightness Adaptation & Discrimination (cont…)



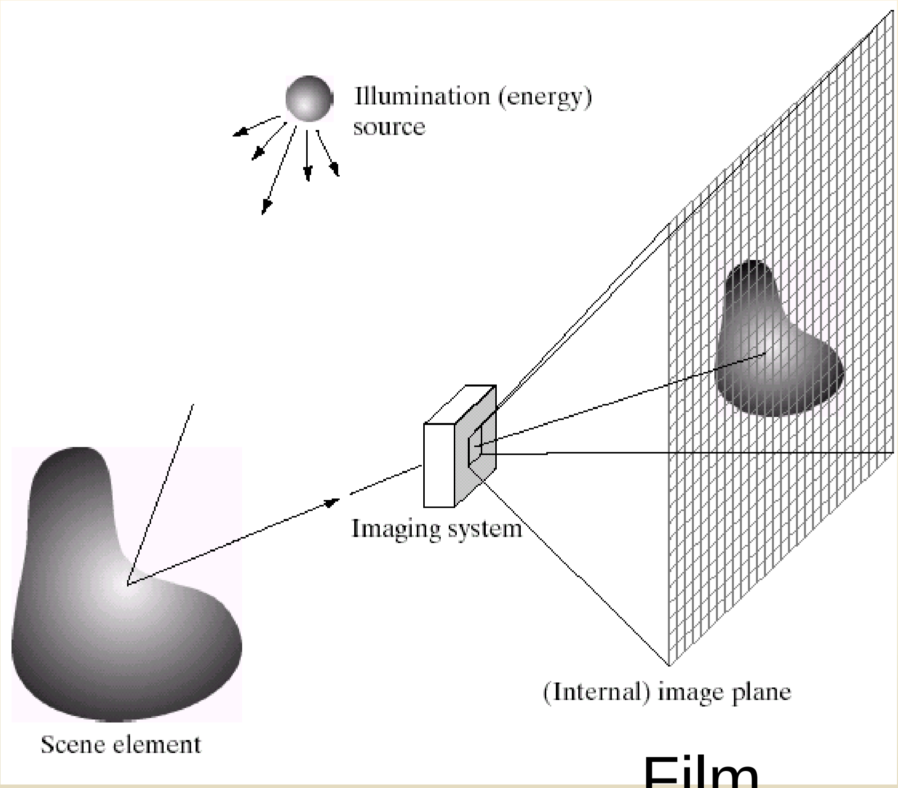
An example of *simultaneous contrast*

# Optical Illusions

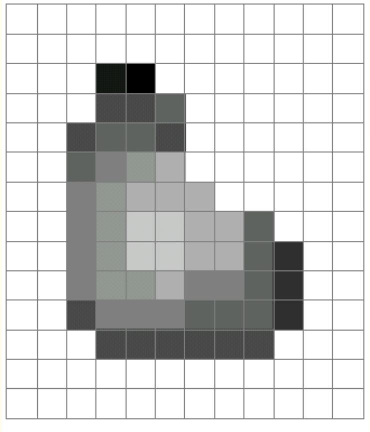


Stare at the cross in the middle of the image and think circles.

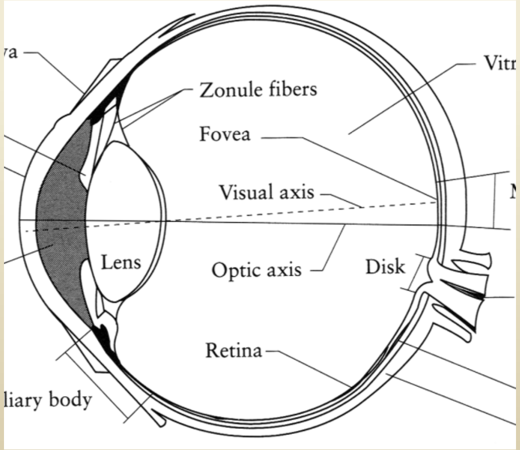
# Image Formation




Film



Digital Camera




The Eye

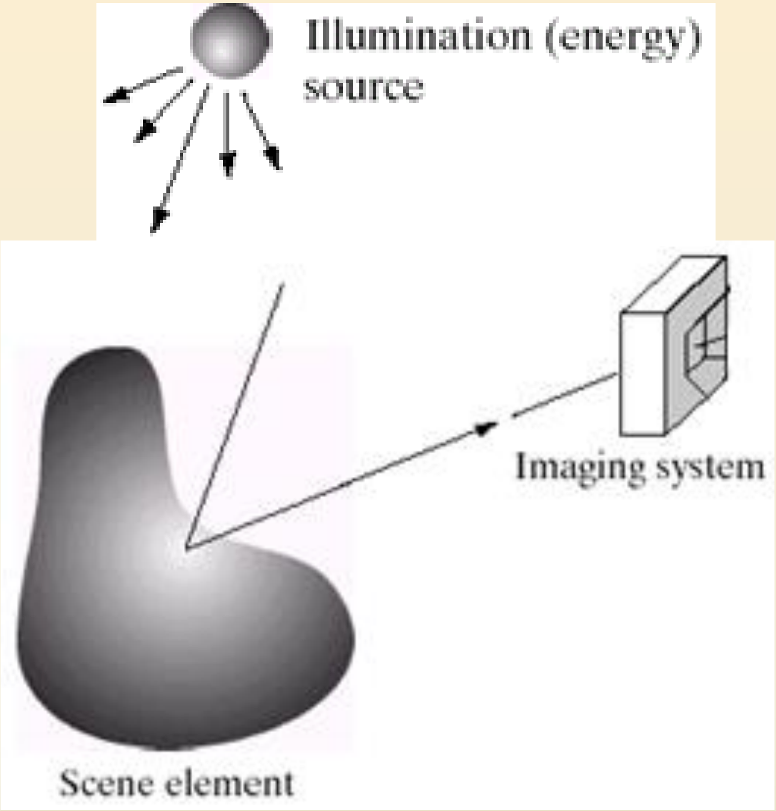
# IMAGE SENSING AND ACQUISITION

# In the following slides we will consider what is involved in capturing a digital image of a real-world scene

- Image sensing and representation and acquisition
- Sampling and quantisation
- Resolution

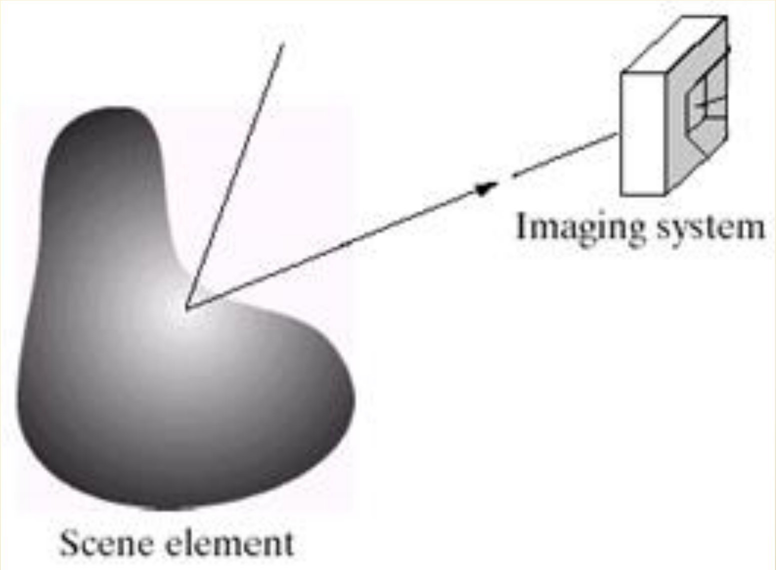
# Image Acquisition

- Three main elements
  1. **Illumination source**
  2. **Scene**
  3. **Sensor (imaging system)**



# Image Acquisition

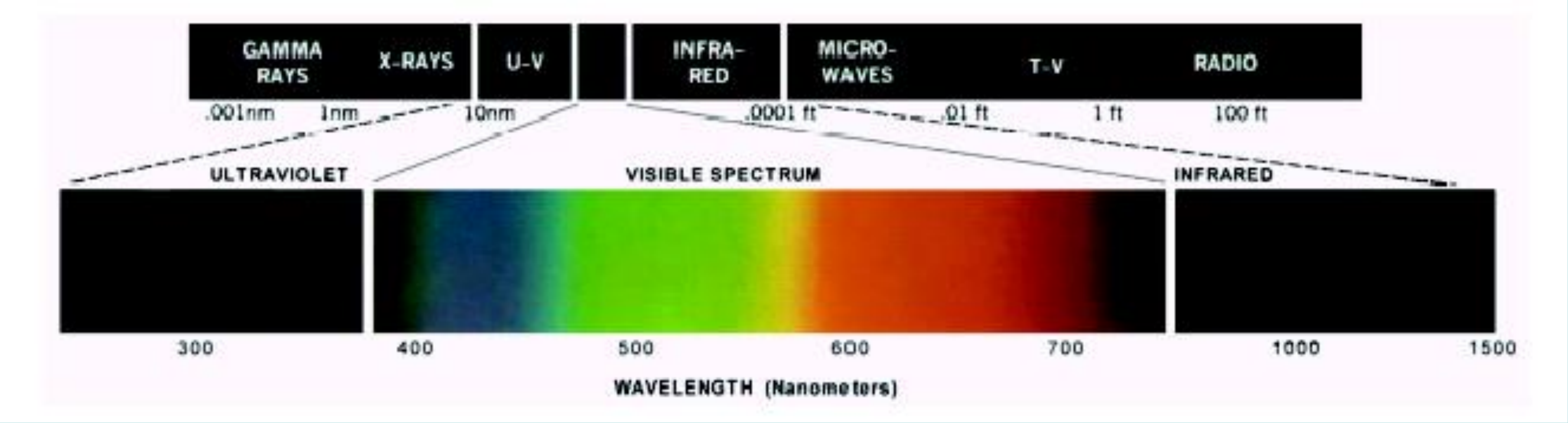
1. **Illumination source:**
   - Can be light energy or
   - EM spectrum
   - Even less tradition sources like
     - Sound, heat


2. **Scene:**
   - Any object: visible or hidden
   - Source itself
3. **Sensor:**
   - Should be capable of sensing the energy

# Light And The Electromagnetic Spectrum

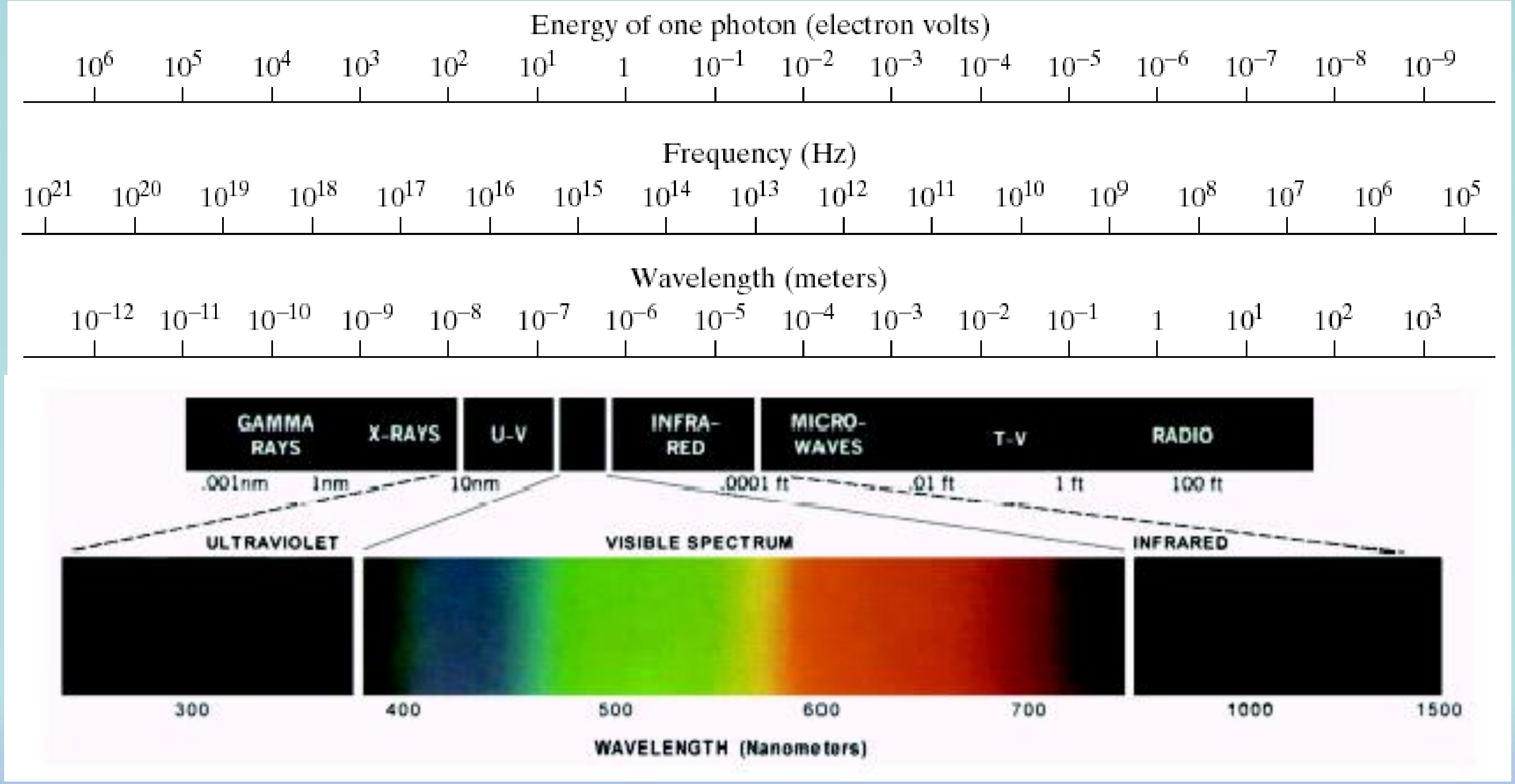
❑In 1666 Sir Isaac Newton discovered that sunlight passed through a prism splits into a continuous spectrum of colors




❑Light is just a particular part of the electromagnetic spectrum that can be sensed by the human eye

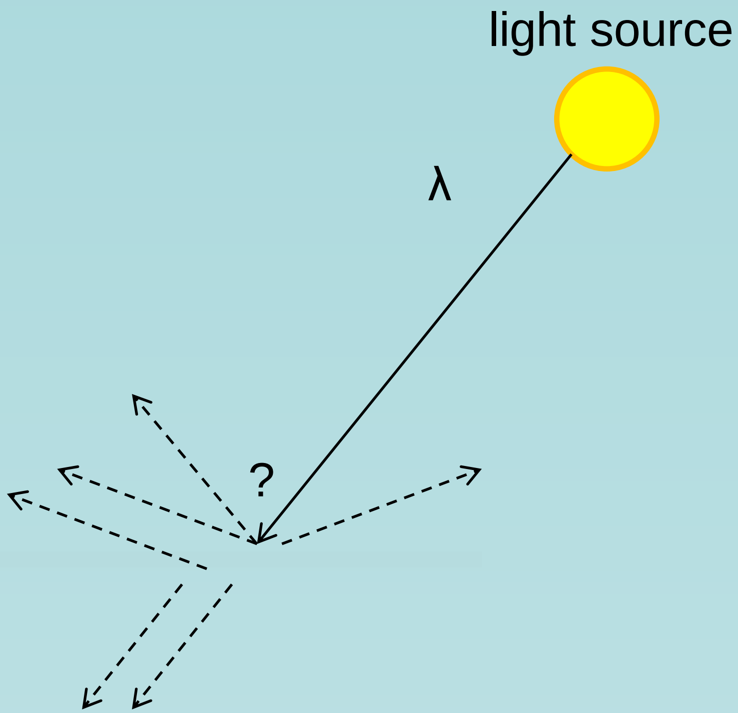
❑The electromagnetic spectrum is split up according to the wavelengths of different forms of energy

- A discrete bundle (or *quantum*) of electromagnetic (or light) energy, PHOTON is proportional to frequency.



# A photon's life choices

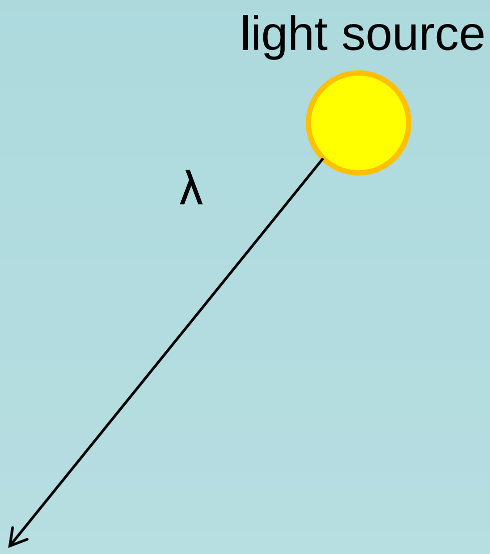
- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection




Chapter 2 of Szeliski

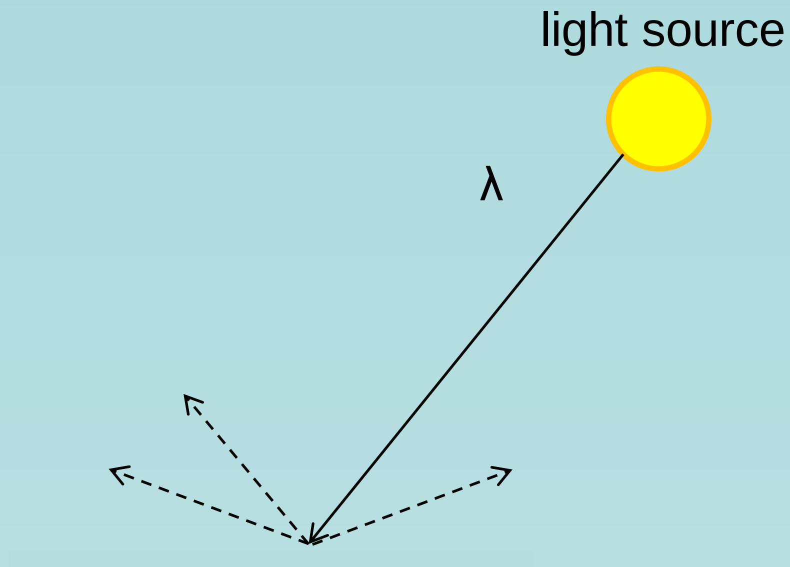
# A photon’s life choices

- **Absorption**
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



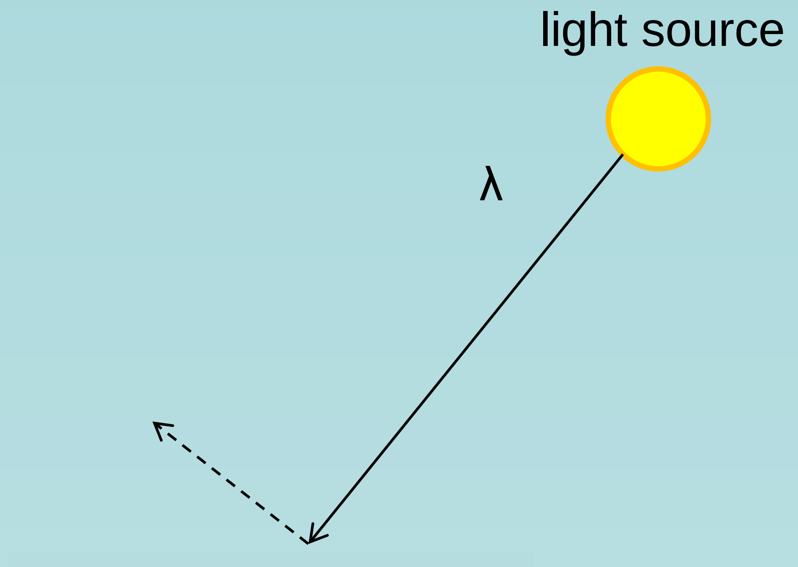
# A photon's life choices

- Absorption
- **Diffuse Reflection**
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



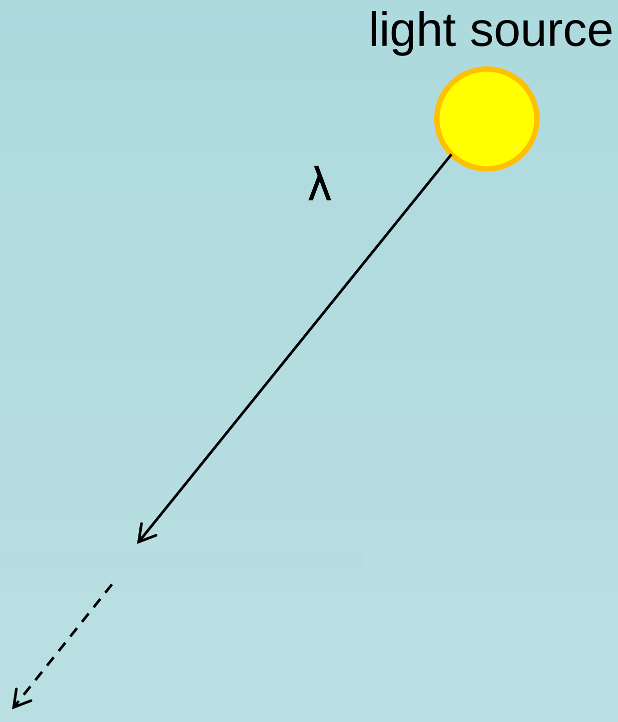
# A photon's life choices

- Absorption
- Diffusion
- **Specular Reflection**
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



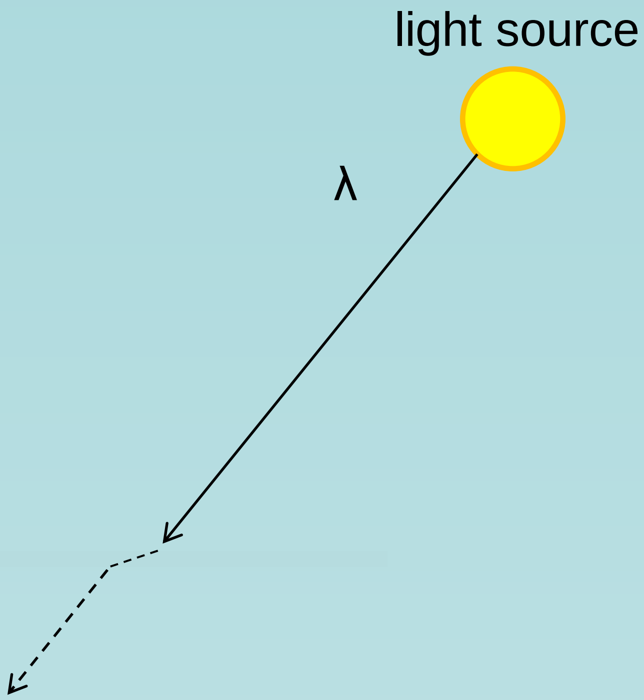
# A photon’s life choices

- Absorption
- Diffusion
- Reflection
- **Transparency**
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



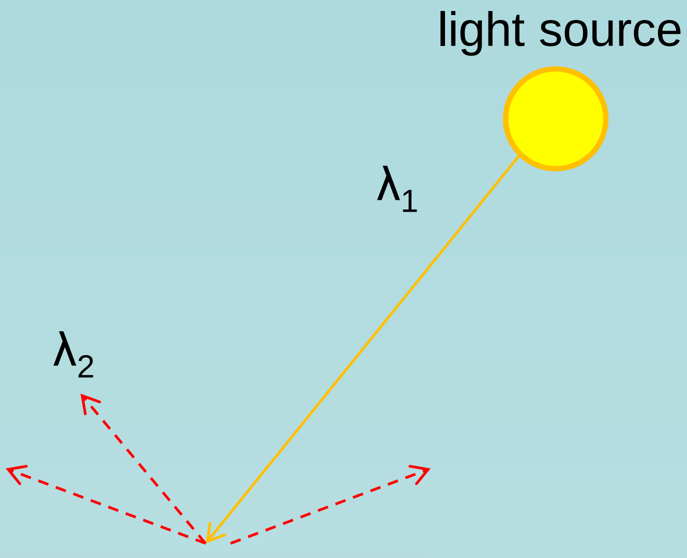
# A photon's life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- **Refraction**
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



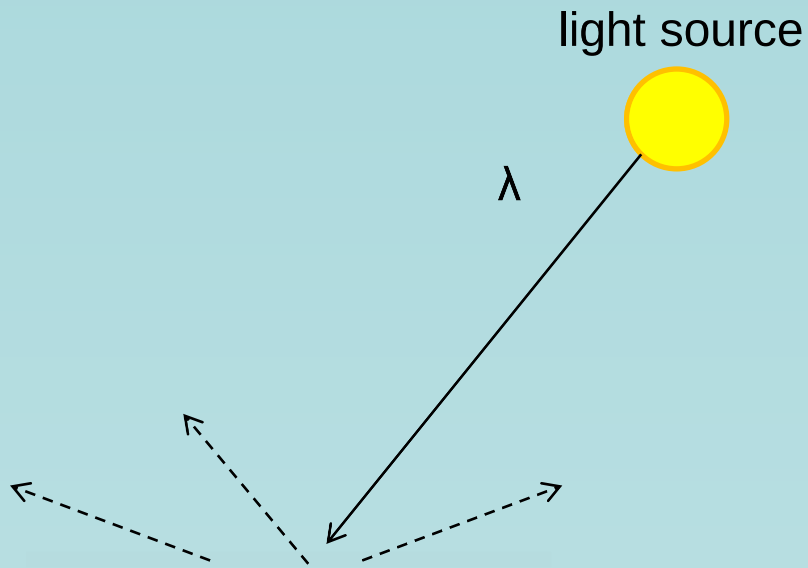
# A photon’s life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- **Fluorescence**
- Subsurface scattering
- Phosphorescence
- Interreflection



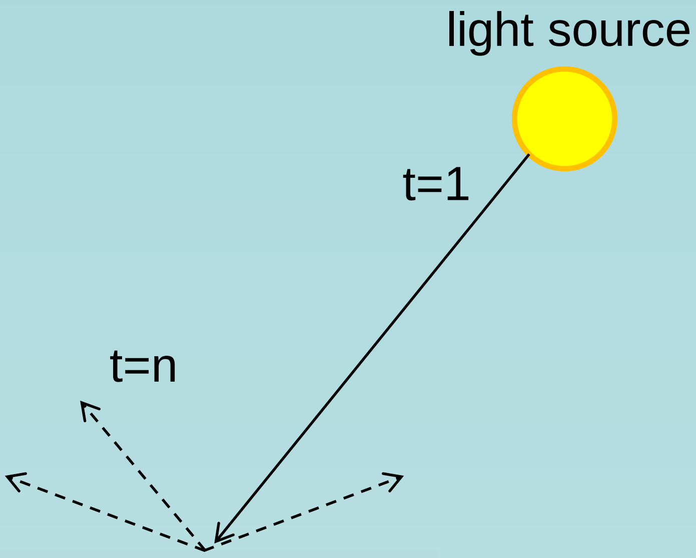
# A photon’s life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- **Subsurface scattering**
- Phosphorescence
- Interreflection



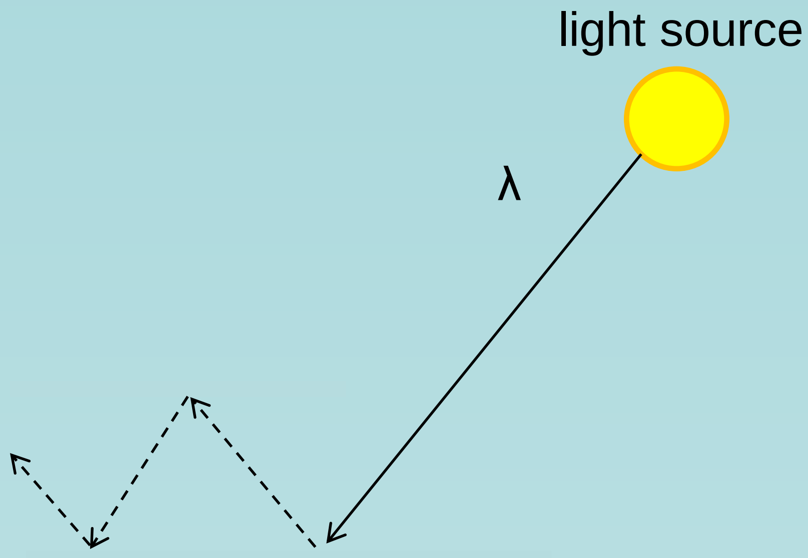
# A photon's life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- **Phosphorescence**
- Interreflection



# A photon’s life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- **Interreflection**

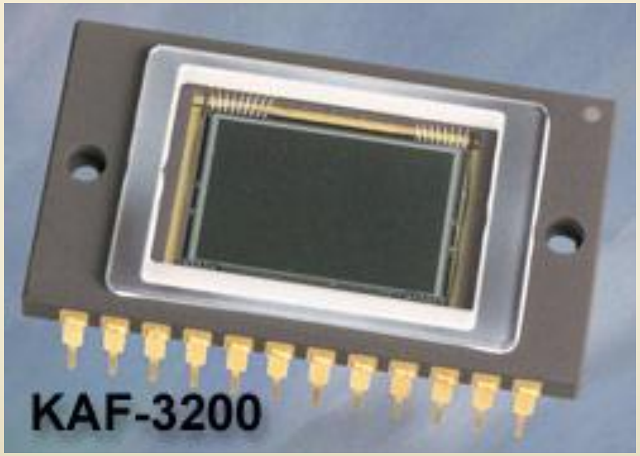



(Specular Interreflection)

# Image Sensors

- ❑ Incoming energy lands on a sensor material responsive to that type of energy and this generates a voltage
- ❑Collections of sensors are arranged to capture images

***Charge-Coupled Device (CCD)***



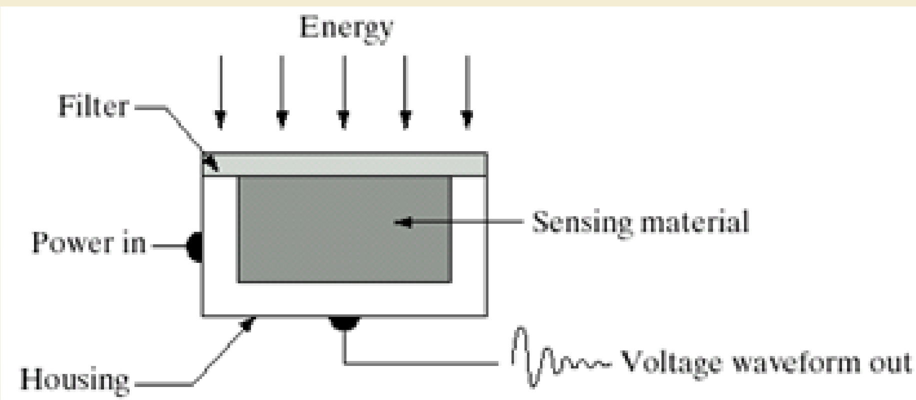

CCD KAF-3200E from Kodak.
(2184 x 1472 pixels,
Pixel size 6.8 microns$^2$)

- ✓ Used for convert a continuous image into a digital image
- ✓ Contains an array of light sensors
- ✓ Converts photon into electric charges accumulated in each sensor unit

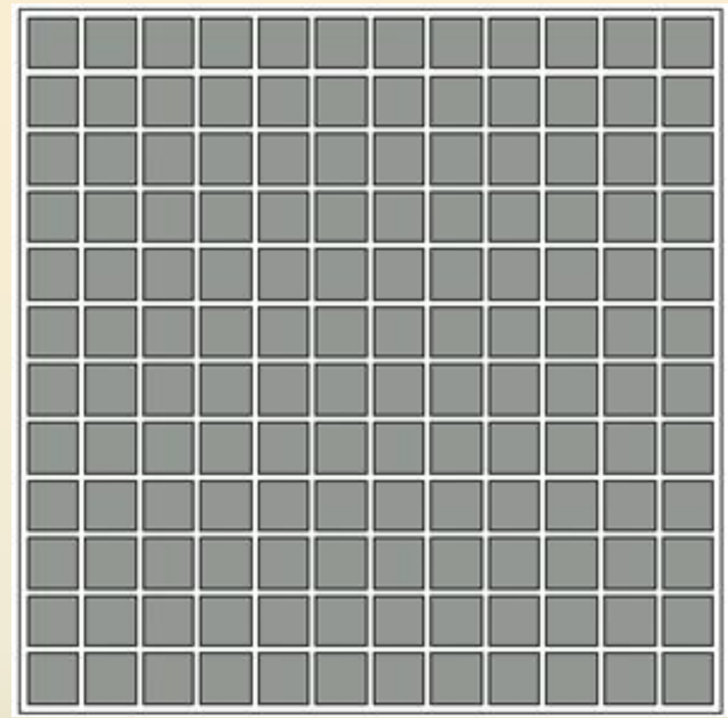
# Image Sensors used to transform illumination energy to digital images.

❑ 3 main sensor arrangements:

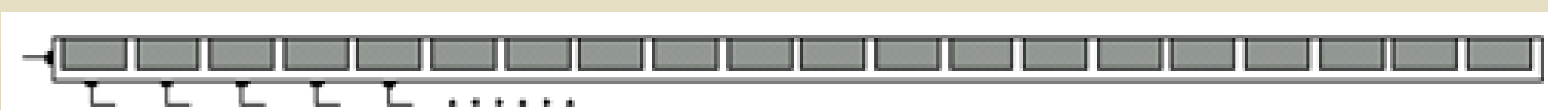
1. **Single sensor**
2. **Line sensor**
3. **Array sensor**




**Single sensor**



**Array sensor**

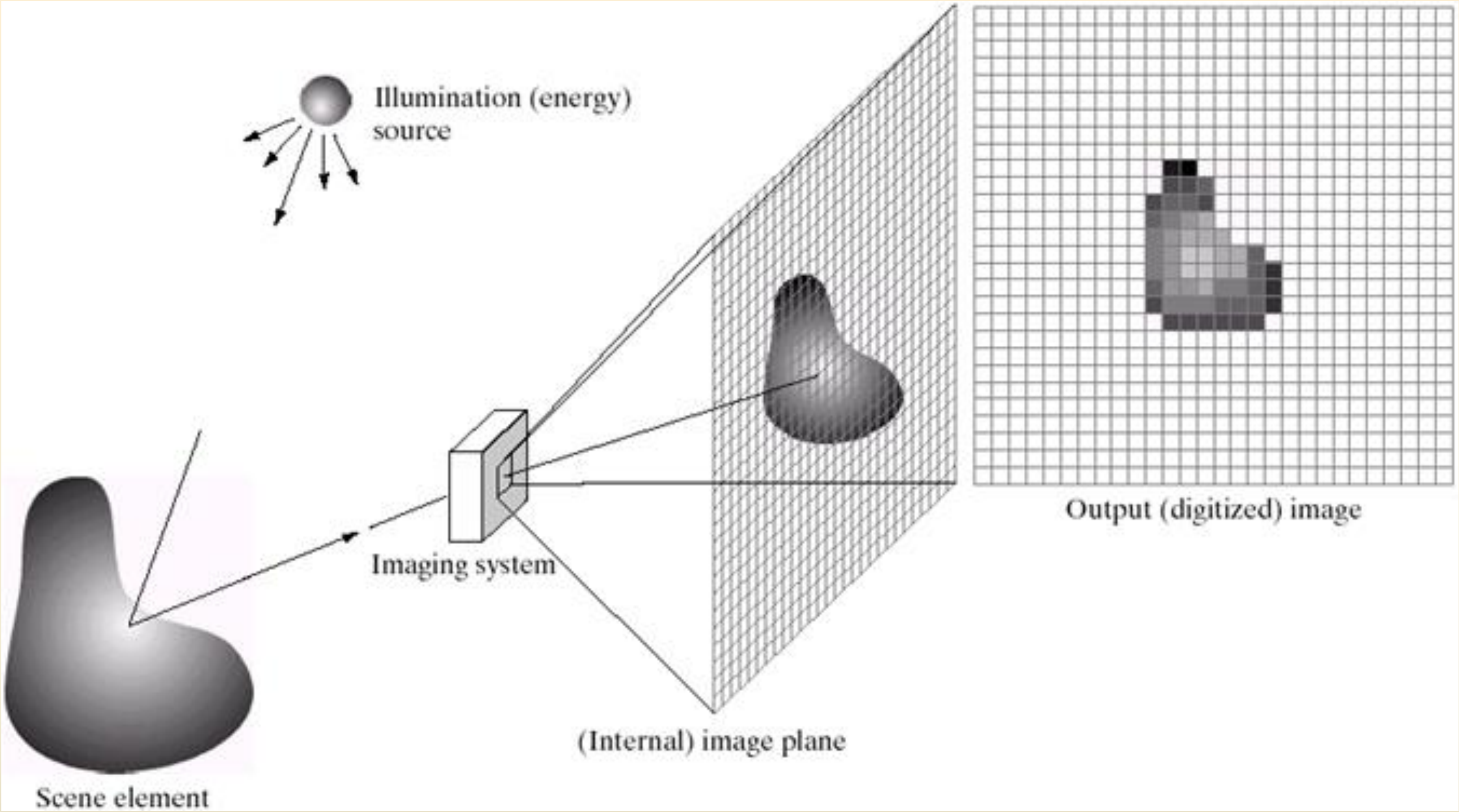


**Line sensor**

❑ Sensing element can be a photodiode

❑ Filter absorbs extra energy or acts as pass-band

# Image Acquisition using Sensor Array




- Images are typically generated by *illuminating* a *scene* and absorbing the energy reflected by the objects in that scene.

# A Simple Image Formation Model

$$f(x, y) = i(x, y) \cdot r(x, y)$$

$f(x, y)$: intensity at the point $(x, y)$

$i(x, y)$: illumination at the point $(x, y)$

(the amount of source illumination incident on the scene)

$r(x, y)$: reflectance/transmissivity at the point $(x, y)$

(the amount of illumination reflected/transmitted by the object)

where $0 < i(x, y) < \infty$ and $0 < r(x, y) < 1$

# Image Formation Model


Image “After snow storm”

$$0 < f(x, y) < \alpha$$

✓**An image**: **a multidimensional function of spatial coordinates**.

✓**Spatial coordinate**: $(x,y)$ for 2D case such as photograph,
$(x,y,z)$ for 3D case such as CT scan images
$(x,y,t)$ for movies

✓The function $f$ may represent intensity (for monochrome images) or color (for color images) or other associated values.

# Some Typical Ranges of illumination

- Illumination

  Lumen — A unit of light flow or luminous flux

  Lumen per square meter ($lm/m^2$) — The metric unit of measure for illuminance of a surface

  - On a clear day, the sun may produce in excess of 90,000 $lm/m^2$ of illumination on the surface of the Earth
  - On a cloudy day, the sun may produce less than 10,000 $lm/m^2$ of illumination on the surface of the Earth
  - On a clear evening, the moon yields about 0.1 $lm/m^2$ of illumination
  - The typical illumination level in a commercial office is about 1000 $lm/m^2$

# Some Typical Ranges of Reflectance

- Reflectance
  - 0.01 for black velvet
  - 0.65 for stainless steel
  - 0.80 for flat-white wall paint
  - 0.90 for silver-plated metal
  - 0.93 for snow

# Human Vision VS Computer Vision


What we see

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 3 | 2 | 5 | 4 | 7 | 6 | 9 | 8 |
| 3 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 2 | 1 | 0 | 3 | 2 | 5 | 4 | 7 | 6 |
| 5 | 2 | 3 | 0 | 1 | 2 | 3 | 4 | 5 |
| 4 | 3 | 2 | 1 | 0 | 3 | 2 | 5 | 4 |
| 7 | 4 | 5 | 2 | 3 | 0 | 1 | 2 | 3 |
| 6 | 5 | 4 | 3 | 2 | 1 | 0 | 3 | 2 |
| 9 | 6 | 7 | 4 | 5 | 2 | 3 | 0 | 1 |
| 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

What a computer sees

# Image Digitization

- Computers use discrete form of the pictures
- The process transforming continuous analog image into discrete approximated image is called digitization.

Analog image → Transforming Process → Digital image

- Two steps:
  - – Sampling and
  - – Quantization

# Image Sampling And Quantisation

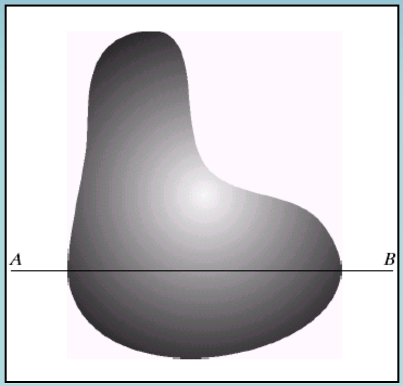
## Sampling:

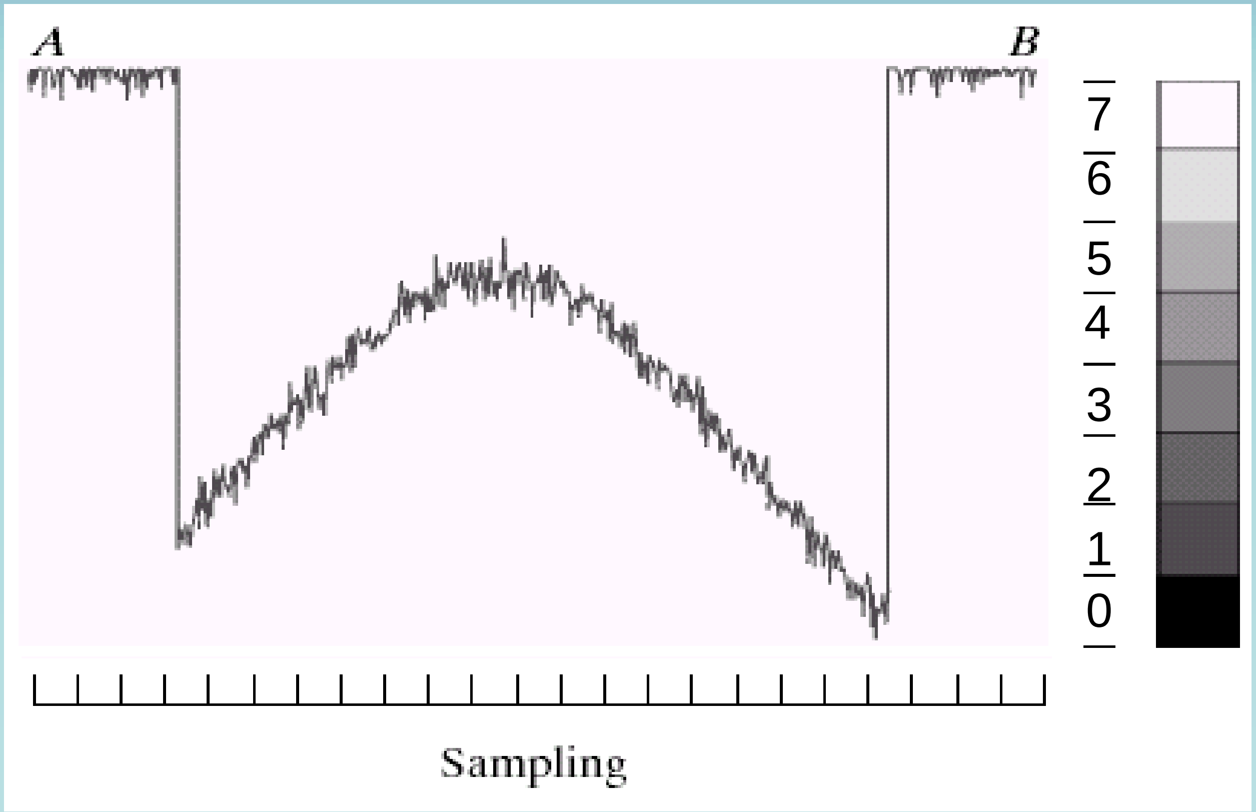
- Digitizing coordinates
- A process which converts the continuous analog space into a discrete space

## Quantization:

- Digitizing amplitudes (gray scale values)
- A process of converting a continuous analogue signal into a digital representation of that signal

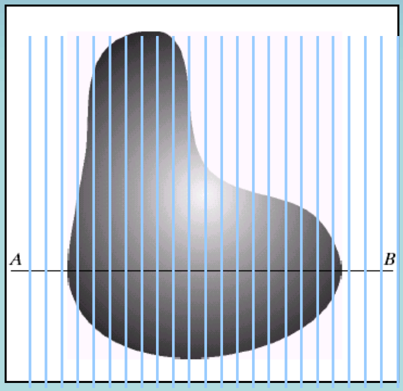
# Image Quantisation

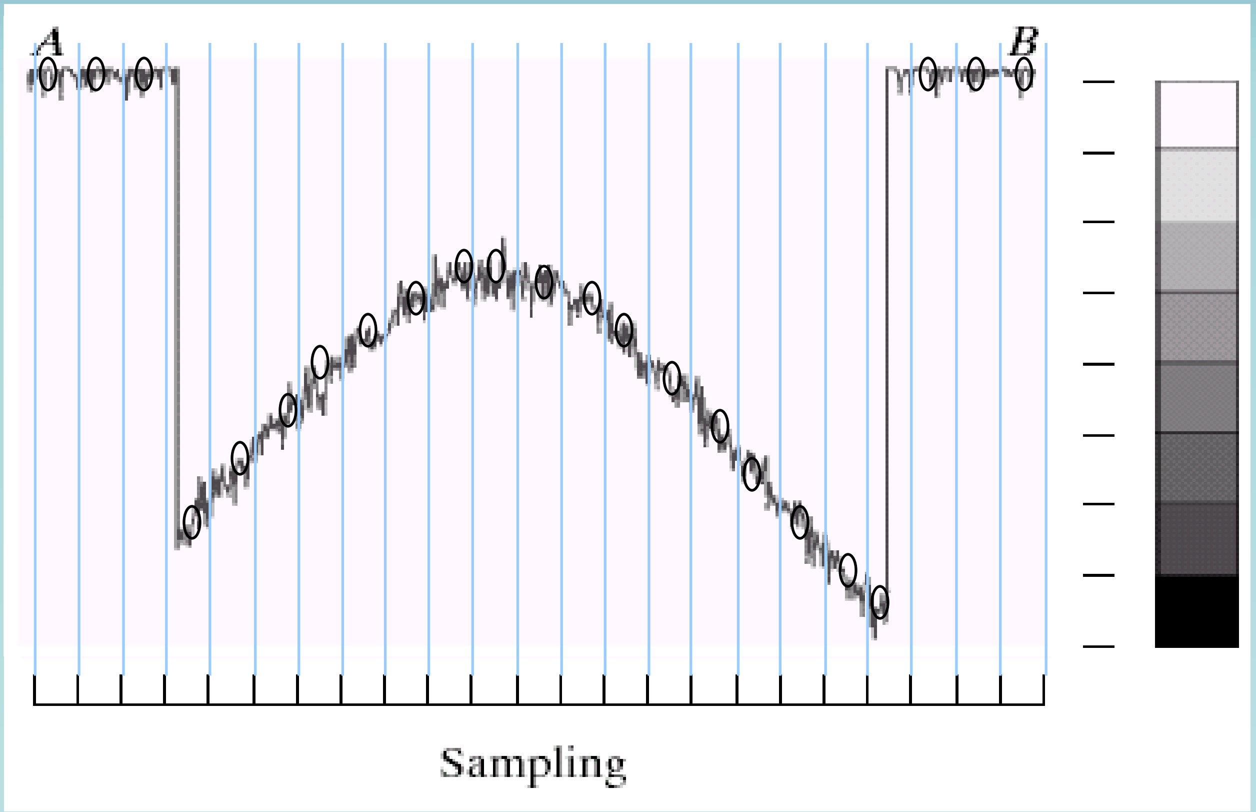



**Digitizing the amplitude values**

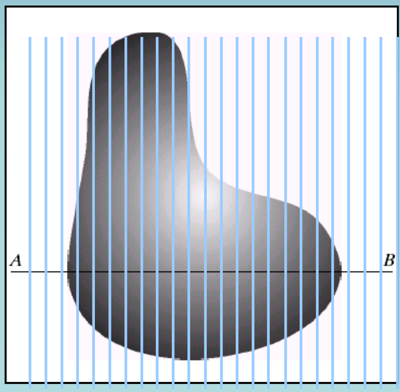
# Image Sampling

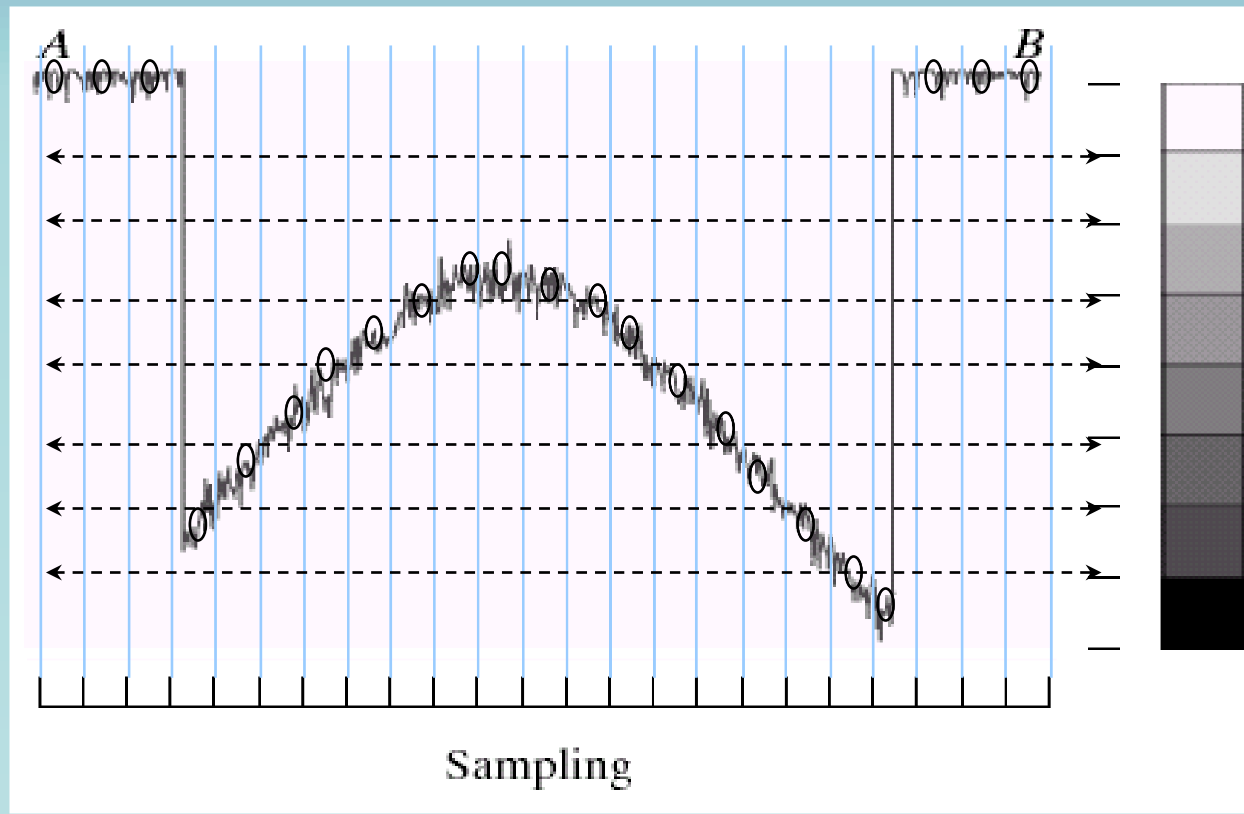



**Digitizing the coordinate values**

# Image Sampling And Quantisation

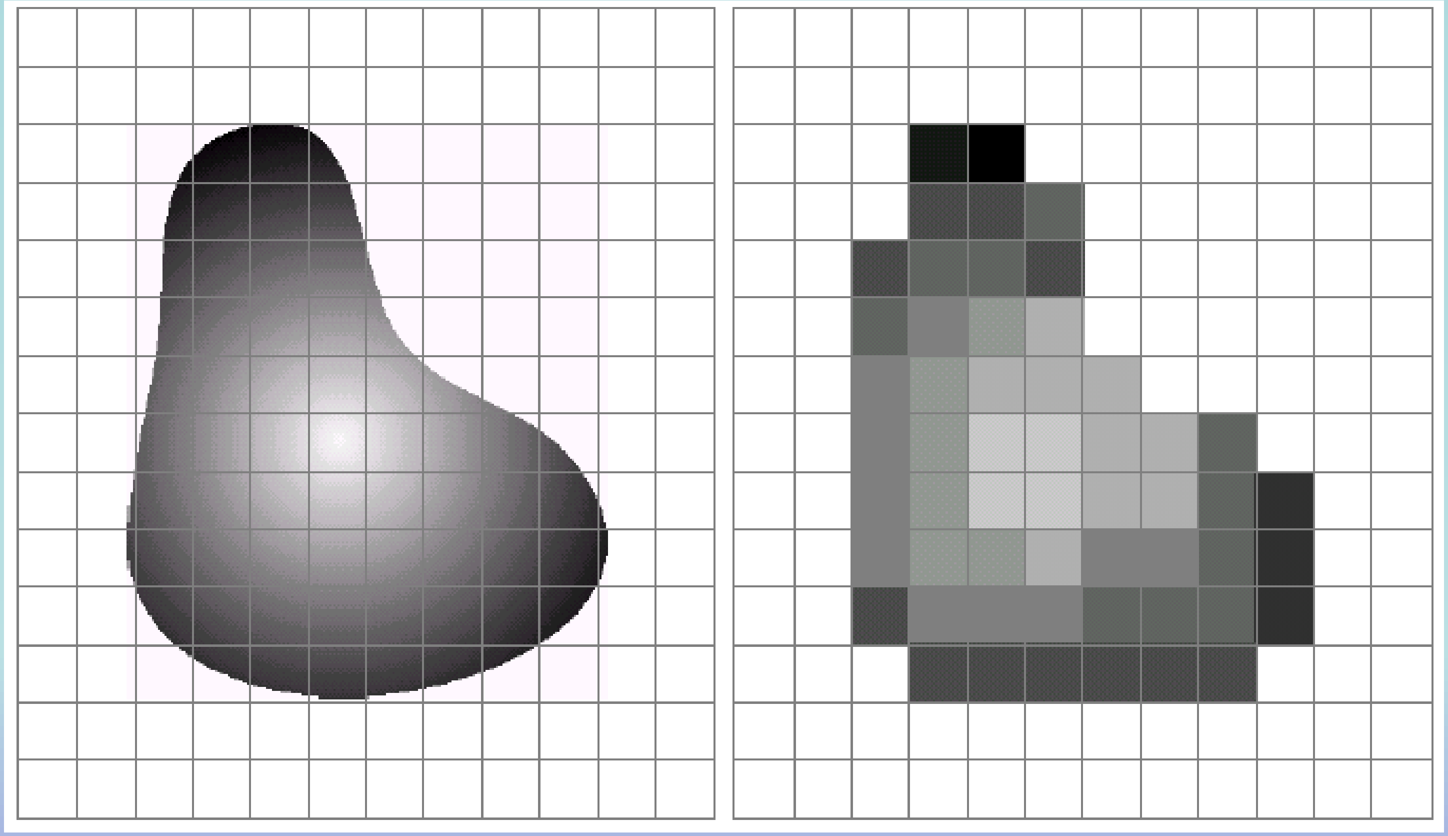


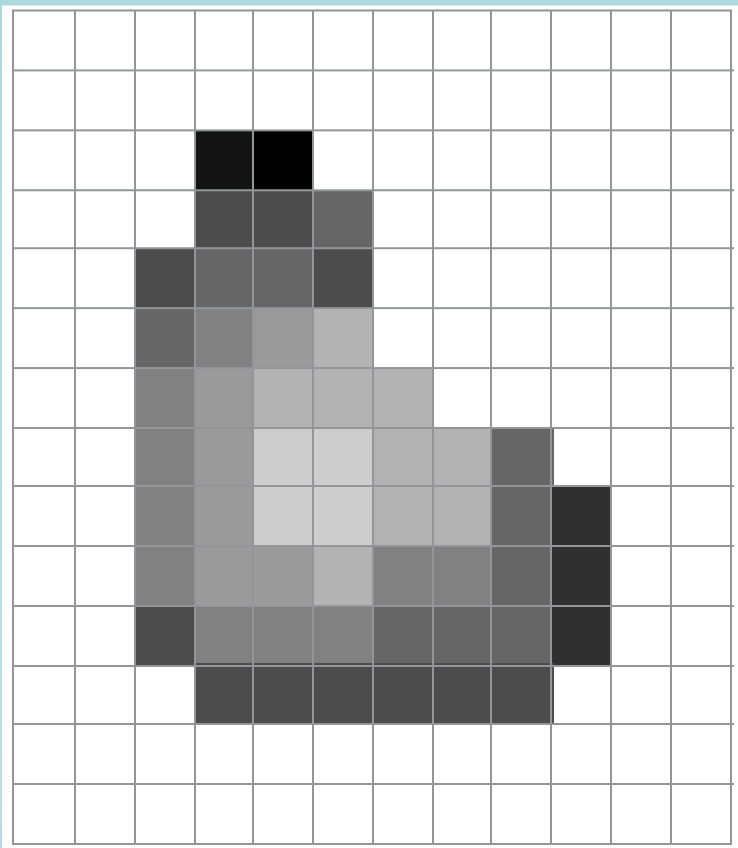

| | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **A** | 7 | 7 | 7 | 1 | 2 | 3 | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 4 | 3 | 3 | 2 | 1 | 0 | 0 | 7 | 7 | 7 | **B** |

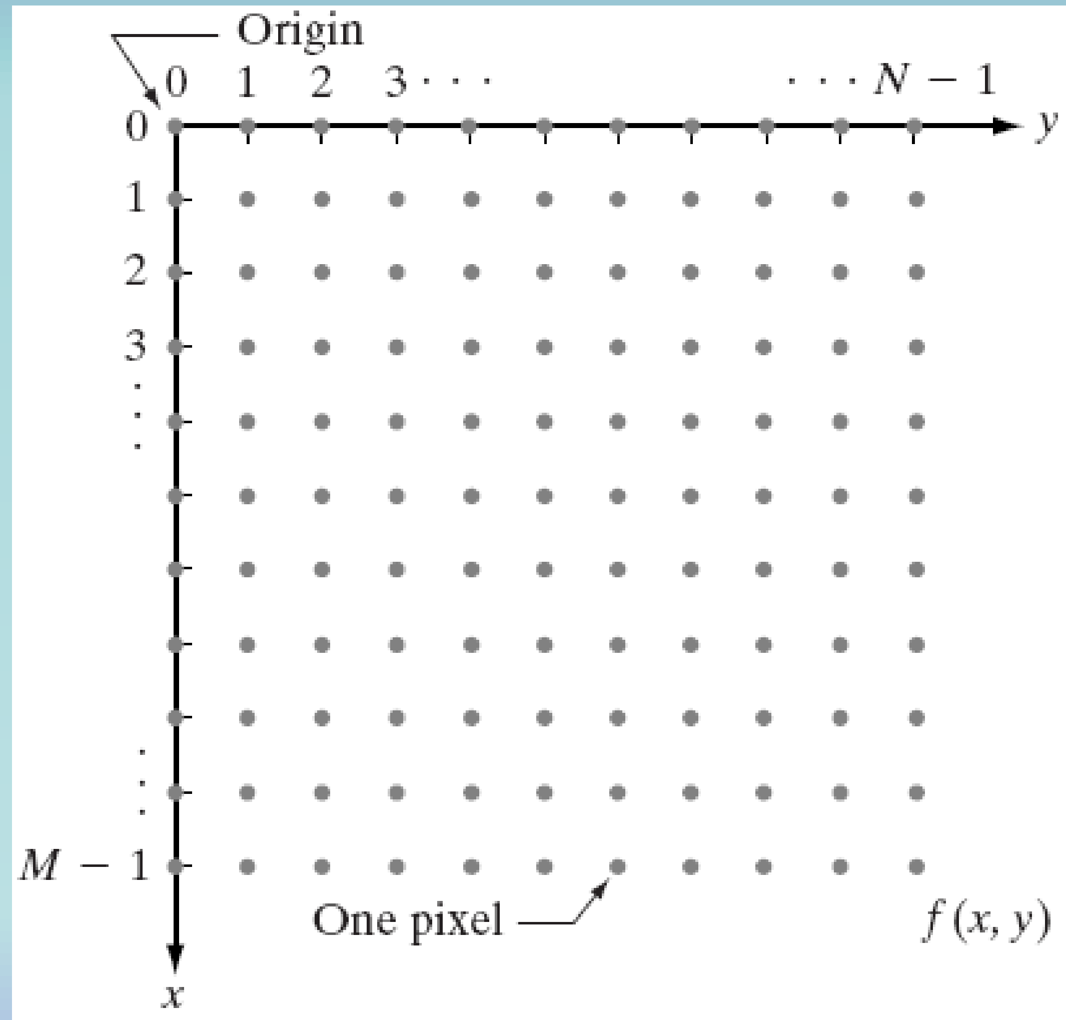
# Image Sampling And Quantisation

- Remember that a digital image is always only an **approximation** of a real world scene



# Representation of Digital Image






Matrix Representation

# Representation of Digital Image

- The representation of an M×N numerical array as

$$f(x,y)=\begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \dots & \dots & \dots & \dots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

# Representation of Digital Image

- The representation of an M×N numerical array as

$$A = \begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,N-1} \\ \cdots & \cdots & \cdots & \cdots \\ a_{M-1,0} & a_{M-1,1} & \cdots & a_{M-1,N-1} \end{bmatrix}$$

**Traditional Matrix**

# Representation of Digital Image

- The representation of an M×N numerical array in MATLAB

$$f(x,y)=\begin{bmatrix} f(1,1) & f(1,2) & \dots & f(1,N) \\ f(2,1) & f(2,2) & \dots & f(2,N) \\ \dots & \dots & \dots & \dots \\ f(M,1) & f(M,2) & \dots & f(M,N) \end{bmatrix}$$

# Next class >>>

- Resolutions
- Types of Digital Image
- Effect of Spatial and Gray Level Resolutions